North South University

Department of Electrical & Computer Engineering

Project Proposal

Submitted to:

Faculty Name: Muhammad Shafayat Oshman

Faculty Initial: MuO

Course Code: CSE299

Course Title: Junior Design

Submitted by:

Name: Ilma Ibrahim

ID: 2321019042

Section: 17

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Problem Statement

With the everyday hustle of life, it is not always easy to keep up with your loved ones. Often times friendships face a "decay" not due to lack of care but due to lack of time and forgetfulness and most of the time it is unintentional. This decay causes the friendship to die at some point and both parties involved may feel helpless in such situations.

Introduction

HoldAString is a hybrid ecosystem designed to strengthen social bonds. The project features a custom prompted AI assistant that works as a Personal Relationship Manager (PRM) and a unique string health system. The string health determines the closeness of the user with a connection and if the health is lower than critical threshold, the AI intervenes to remind the user to check-in. Moreover, the AI remembers key details from the user's last hangout or interactions with a connection. By doing so, the AI holds the ability to provide context aware nudges to ensure that every check-in is meaningful.

Project Description

The project consists of a web portal and a dedicated Android application. The website handles the primary features and the mobile app is hybrid, with an additional feature where the users can complete mini games with their connections to gain additional string health, given that the connection is a user as well.

The backend of the project is primarily based on PHP, while mySQL is used for the database. We use cURL to integrate the API to the software and send data to the Google API. To test the AI prompts and database responses, we use K6 tool. For email verification, we use the PHPMailer library.

The frontend is built using HTML, CSS, and JavaScript. For the mobile app, we use WebView to run the PHP/HTML codes. Java will be used as the specific language for the .apk.

Planned Features

- **The Primary PRM Chat**

  The user writes a brief text or records a 2 minutes voice message of their most recent hangout with the connection. The Google API extracts the key details and stores them into the database, and then provides context-aware nudges and advice in reference to past interactions.

  Example of the details extracted and stored by the AI –

  Key Person (The connection): Mary

  Event: Afternoon Hangout on 12-02-2026
  Antagonist(s) (If any): Luna (Mary's cat)
  Upcoming Event(s): Luna's surgery (Next Wednesday)

Follow-up Topic(s): Luna's health
Suggested Question(s):
How did the surgery go?
How is Luna doing?

- **Individual Connection Profiles**

  This features as a dedicated dashboard for each individual string.

- **String Health Monitoring**

  The string health is calculated based on interaction frequency. The data required for this calculation is primarily fetched from the "isResolved" boolean factor in the Events table and "lastInteraction" timestamp in the Strings table. A background PHP script calculates a decay rate for the string health; for example, a reduction of 5 health for every 24 hours of inactivity with the connection. If the string health is below 50, the AI gives the user a nudge.

- **Duo Mini-games (App Exclusive)**

  This additional feature is exclusive for the mobile app to incentivize active real-time engagement. There will be minigames that the user can play with their connection if they're connected through the app. The completion of a minigame provides an instant +10 boost to the string's health. The app will feature one or multiple minigames, given depending on how much time we have left by the end.

Weekly Breakdown of Project Plan

Week 1: Project Idea Submission
Week 2: Database creation, user registration and login implementation
Week 3: Connecting the API; designing and testing system instructions for the AI
Week 4: CRUD implementation
Week 5: Construction of the User Interface
Week 6: Building the connection with the app and implementing app-only features (Part 1)
Week 7: Implementing app-only features (Part 2)
Week 8: Building a proper email verification process
Week 9: Refinement of the User Interface where necessary

Conclusion

HoldAString is designed to help users maintain meaningful human connections. Its system proactively reminds user of important life events —whether it is the birthday or an important job interview of a friend —that often get lost in daily life. The overall project demonstrates how a hybrid mobile architecture and artificial intelligence can be used together to ensure that no important detail in a relationship is ever forgotten.